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GRAPHIC REPRESENTATION OF PAIN

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SUMMARY

Of the different types of visual analogue and graphic rating scales tested in a series of experiments, only two were satisfactory: these were the visual analogue scale and the graphic rating scales used horizontally with the words spread out along the whole length of the line. Other types of scale used gave distributions of results which were not uniform. Unusual distribution of results occurred when patients selected a position adjacent either to descriptive terms or preferred numbers. In some experiments, the distribution of results was determined by the nature of the experiment. Alternation of the ends of a scale did not affect the results. The behaviour of the graphic rating scale was different in patients accustomed to completing it and in those not so accustomed.

The results of pain severity measured by these methods showed a very good correlation with pain severity measured by the simple descriptive pain scale. Changes in visual analogue scores also correlated well with changes in simple descriptive pain scores. The visual analogue and graphic rating scales were more sensitive than the traditional simple descriptive pain scale. Most patients could readily use visual analogue and graphic rating scales despite having no previous experience. The failure rate was slightly lower with the graphic rating method. Use of these scales is the best available method for measuring pain or pain relief.

INTRODUCTION

Visual analogue scales have proved to be satisfactory in the subjective measurement of pain. A visual analogue scale is a straight line, the ends of which are defined as the extreme limits of the sensation or response to be measured. A graphic rating scale is a visual analogue scale with descriptive terms placed at intervals along the line, and such a scale might have advantages for the patient who is required to use it. The descriptions might assist the patient in deciding the position of his score (especially if he has no experience of pain measurement) and make different patients more likely to record the same degree of severity in the same position. The result of experiments involving change might be more understood by physicians; for example, a slight improvement has more meaning than a score change from one number to another.

In this paper a series of different types of visual analogue and graphic rating scales have been studied.

METHODS

Six different types of visual analogue and graphic rating scales have been tested. Each scale was tested in 100 consecutive out-patients with painful

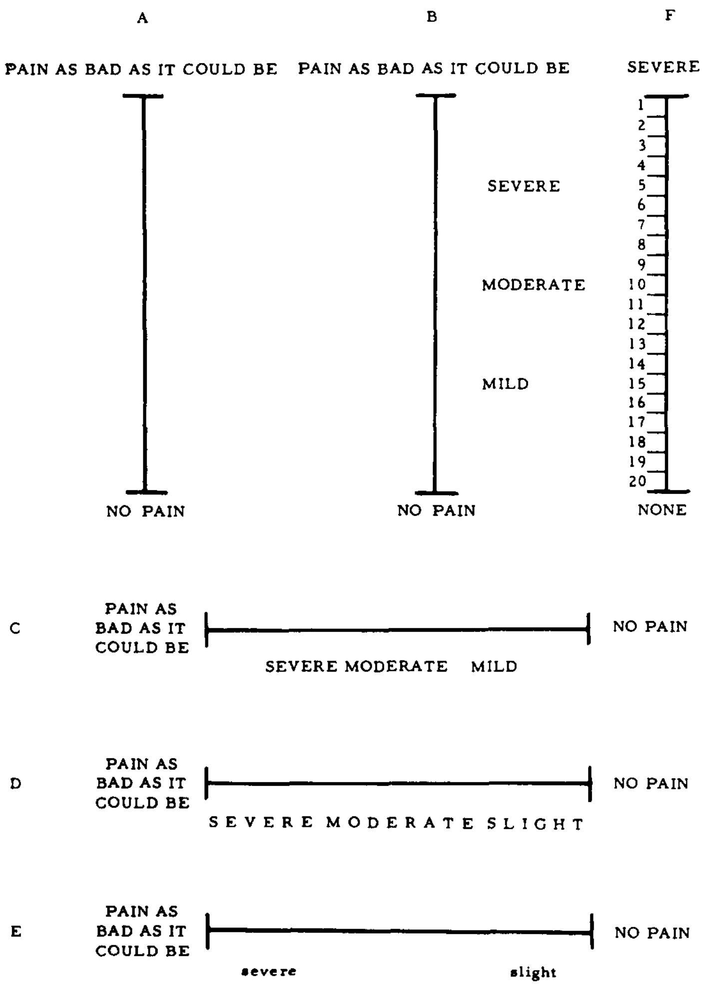

Fig. 1. Visual analogue and graphic rating scales used in the study.

conditions, none of whom had seen a visual analogue or graphic rating scale before. One observer gave a standard explanation of the method to all the patients and allowed up to 30 sec for completion of the scale. They were considered to be failures if they had not used the scale within this time, or completed it in a manner which was not interpretable.

Experiment I

A vertical visual analogue scale (Fig. 1A) was used at the same time as the simple descriptive pain scale described by Keele [6].

Experiment II

A vertical graphic rating scale (Fig. 1B) was used, with the descriptions severe, moderate and mild at equal intervals along the line. A simple descriptive pain scale was also used.

Experiment III

A horizontal graphic rating scale (Fig. 1C) was used with the same words, their middle points being in the same positions as in the previous experiment.

Experiment IV

A horizontal graphic rating scale (Fig. 1D) was used with the descriptions

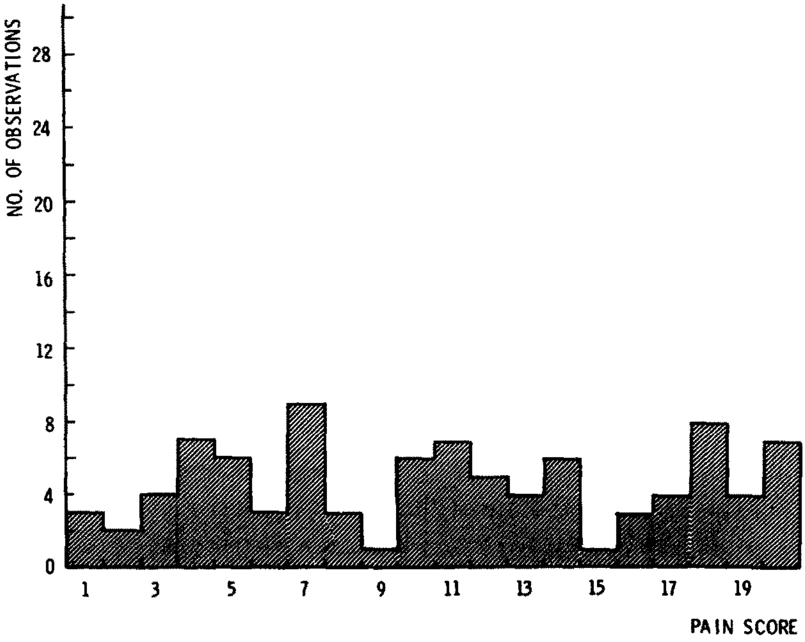

Fig. 2. Uniform distribution of results obtained from the use of a visual analogue scale (Fig. 1A).

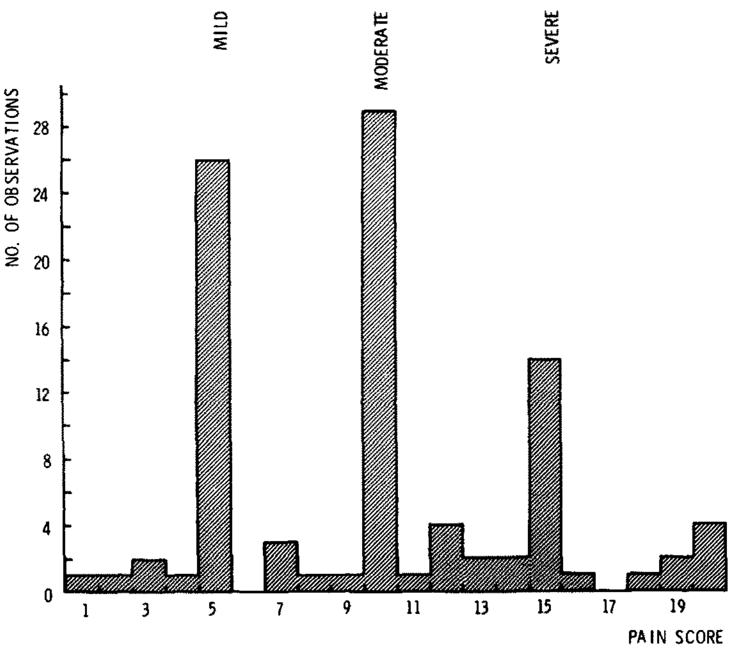

Fig. 3. Non-uniform distribution of results obtained from the use of a vertical graphic rating scale (Fig. 1B).

severe, moderate and slight. These words were spread along the entire length of the line. The extreme ends of the scale were alternated, 50 patients using a scale in which severe pain was at the left end and 50 patients using a scale in which severe pain was at the right end.

Experiment V

A horizontal graphic rating scale (Fig. 1E) was used, with severe and slight being the only descriptions.

Experiment VIa

A vertical graphic rating scale (Fig. 1F) was used, but calibrated with numbers from 1 to 20 instead of descriptions.

Experiment VIb

The same scale as in the previous experiment was tested in 100 out-patients who were experienced in the use of this type of scale.

Experiment VII

During a long-term trial of penicillamine and gold in 86 patients with rheumatoid arthritis, both visual analogue and simple descriptive pain scales were used. Both scales were used before the start of treatment and 6 months later.

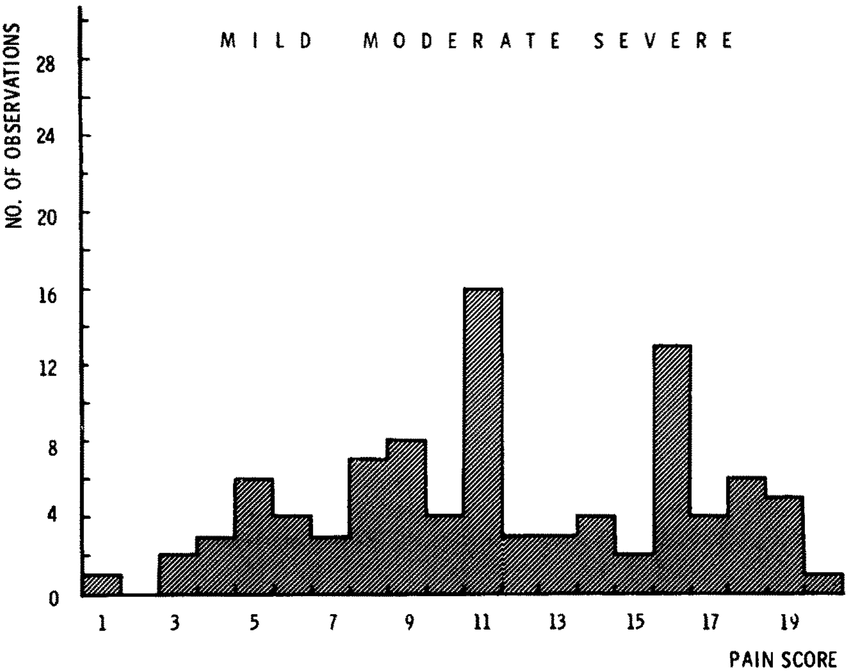

Fig. 4. Non-uniform distribution of results obtained from the use of a vertical graphic rating scale (Fig. 1C).

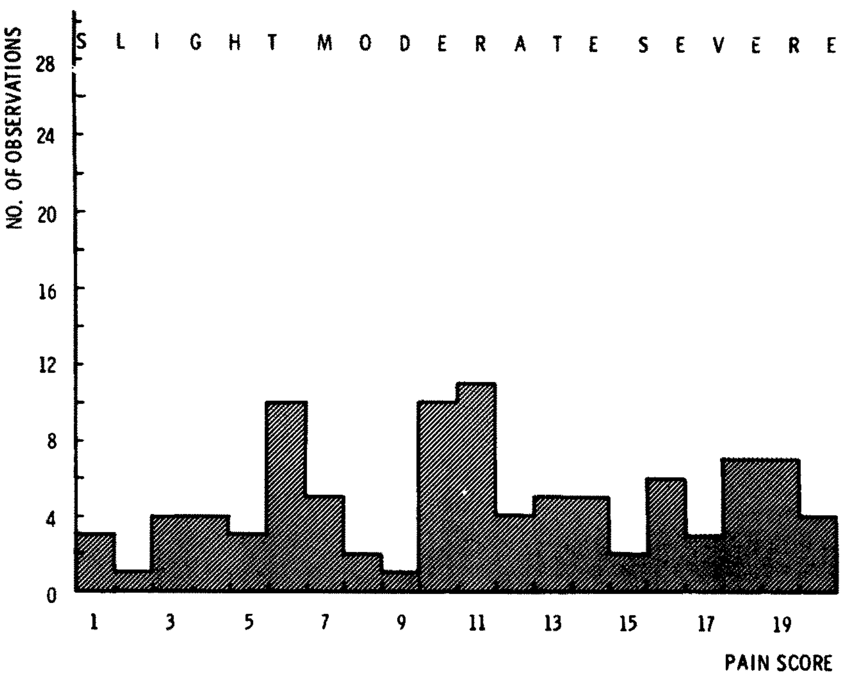

Fig. 5. Uniform distribution of results obtained from the use of a vertical graphic rating scale (Fig. 1D).

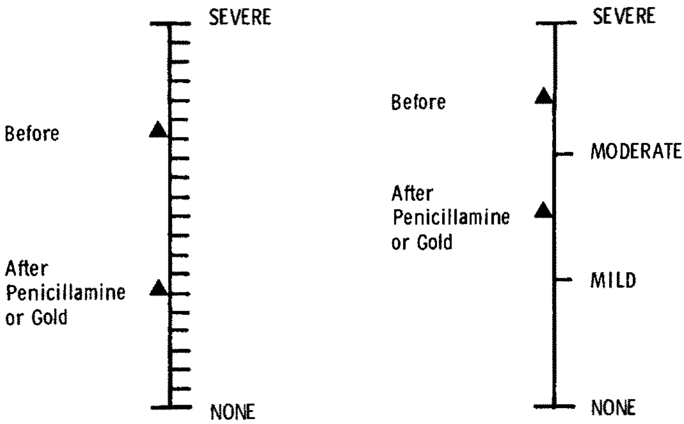

Fig. 6. Effects of 6 months treatment with penicillamine or gold on mean pain scores, using simultaneously a visual analogue scale and a simple descriptive pain scale (experiment VII), showing the increase in sensitivity obtained from the visual analogue scale.

Experiment VIII

Visual analogue scales were used in a trial of knee replacements. The patients were asked to complete a scale in a pre-operative assessment and in post-operative assessments at 6 monthly intervals.

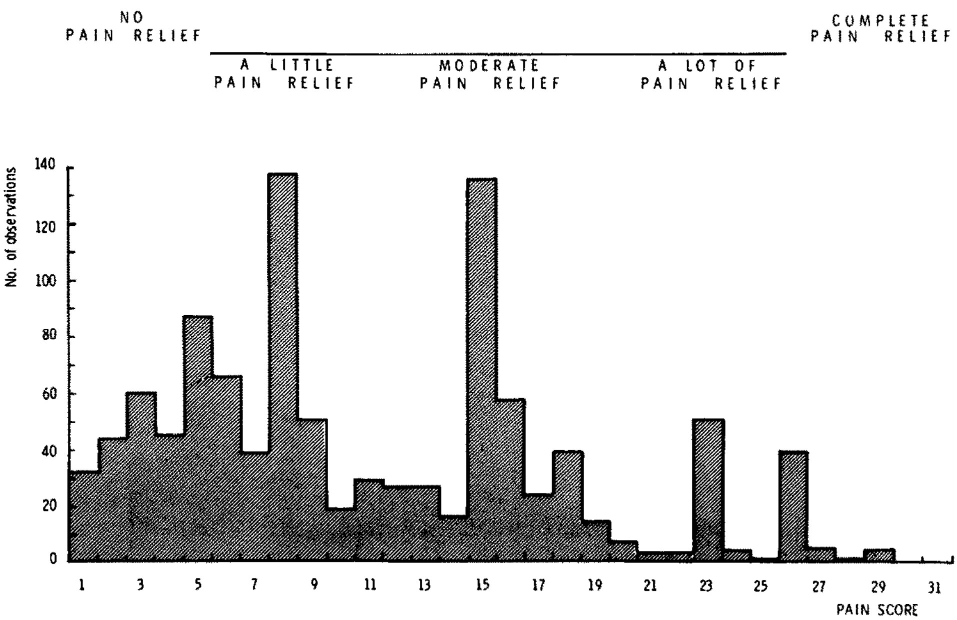

Fig. 7. Graphic rating pain relief scale used in an analgesic study and the non-uniform distribution of results obtained from it.

Experiment IX

Twenty-four patients were admitted to a trial of single doses of a new analgesic compound. Each patient was given 3 doses of the new compound, a standard analgesic and placebo (a total of 9 doses). These were given on consecutive days in a randomised and balanced design. Patients were asked to complete a graphic rating pain relief scale for each of the 6 consecutive hours after the administration of the test drugs. The ends of the line were defined as 'complete pain relief' and 'no pain relief'. The descriptions 'a little pain relief', 'moderate pain relief' and 'a lot of pain relief' were placed at equal intervals along the line but the ends of the line had no 'cut off' point (Fig. 7). Patients were given a standard explanation of the scale by the same observer.

STATISTICAL METHODS

The scale was divided into 20 portions by the use of a stencil applied at the end of each experiment. To test the uniformity of the distribution of results, adjacent categories were combined to form 10 classes and the chi-squared test applied. Correlation coefficients were used to examine the relationship between different measurements.

RESULTS

Experiment I

The distribution of the scores did not differ significantly from a uniform distribution (χ^2 = 8.61, $P > 0.1$) (Fig. 2). There was a statistically highly significant correlation with the simple descriptive pain scale ($r = 0.75$, $P < 0.01$). Seven percent of patients failed to complete the scale, mainly being unable to understand the concept.

Experiment II

The distribution of the scores differed significantly from a uniform distribution (χ^2 = 102.75, $P < 0.01$) (Fig. 3). Seventy-three percent of the results were clumped near the descriptive terms and only 27% of patients achieved the increased sensitivity by placing their marks at points between the descriptive terms. The correlation with the simple descriptive pain scale remained extremely high ($r = 0.9$, $P < 0.01$), and the failure rate was reduced to 4%.

Experiment III

The distribution of the scores differed significantly from a uniform distribution (χ^2 = 25.11, $P < 0.01$) (Fig. 4). Though only 55% of the line had descriptions against it, 78% of patients used the descriptions. The failure rate was similar at 5%.

Experiment IV

The distribution of the scores did not differ significantly from a uniform distribution ($\chi^2 = 9.08, P > 0.1$) and there was no tendency for the scores to be placed against the descriptive terms (Fig. 5). There was no difference between the distribution of the two sets of scores from alternating the extreme ends of the scale ($\chi^2 = 8.93, P > 0.1$). The failure rate was 3%.

Experiment V

The distribution of the scores differed significantly from a uniform distribution ($\chi^2 = 21.3, P < 0.01$). There are peaks over the words slight and severe, and another at the middle point. The failure rate was 2%.

Experiment VIa

The distribution of the scores differed significantly from a uniform distribution ($\chi^2 = 31.92, P < 0.01$) with 10 and 15 being favoured numbers.

Experiment VIb

The distribution of the scores did not differ significantly from a uniform distribution ($\chi^2 = 12, P < 0.1$), though there were slight peaks over the numbers 10 and 20. The majority of these patients had much experience of these scales and did not use the numbers; indeed, many did not even notice them.

Experiment VII

There was a statistically highly significant correlation between the change in pain scores using the visual analogue pain scale with the change scores using the simple descriptive pain scale after 6 months treatment ($r = 0.6, P < 0.01$). There were also good correlations between changes in the pain scale using the visual analogue scale and changes in some other measures of disease severity such as articular index, duration of morning stiffness, grip strength and progress (measured on a 5-point scale). The changes measured on the two scales are shown in Fig. 6.

Experiment VIII

This trial demonstrated how odd distributions could be obtained in an individual experiment simply because of the nature of the experiment. Before operation, these patients had severe or moderate pain, but one year after operation, the majority had little pain. The before and after operation pain scores therefore gave a U-shaped distribution.

Experiment IX

Twenty-one patients completed the experiment. The results are clumped near the descriptive terms and the distribution of the scores differed significantly from a uniform distribution ($\chi^2 = 732.17, P < 0.01$) (Fig. 7). Many patients took the ends of the actual line as the limits of the possible amount of pain relief, but some patients took the ends of the defining descriptive terms, and others took the middle points of these words as their limits of pain relief.

DISCUSSION

This study has shown that some types of visual analogue and graphic rating scales are satisfactory while others are not; minor changes may alter the behaviour of a scale and should therefore be avoided. The advantage of scales of this type is their sensitivity — the lack of sensitivity of the simple descriptive pain scale has been demonstrated by Huskisson et al. [5] who showed that the mean change produced in a group of patients by anti-inflammatory drugs and placebo took place entirely within one grade of the scale. A ruler graduated in centimetres cannot measure millimetres any more than the simple descriptive pain scale can measure the small changes produced by analgesics and anti-inflammatory drugs. The use of a visual analogue as well as a simple descriptive pain scale in a group of patients treated with penicillamine demonstrated the greater sensitivity of the visual analogue scale. The sensitivity of a measurement is defined as the number of units of change for a given force applied. The force applied to a pain scale is the improvement produced by a particular treatment which cannot be measured in exact terms. However, the same force was applied to both scales in this experiment and whereas the simple descriptive pain scale changed one unit, the visual analogue scale changed 8 units. It is therefore reasonable to conclude that the visual analogue scale was more sensitive in this particular experiment. Though the use of more than 20 divisions on a visual analogue scale would lend to an apparent increase in sensitivity, there must be a limit to the size of the differences that patients can discriminate. The use of 20 divisions receives some support from the work of Hardy et al. [2] who found 21 just noticeable differences between the perception of pain and the limit of tolerance. Sensitivity of visual analogue and graphic rating scales is reduced when the distribution is not uniform. If all the results were clumped against descriptive terms, the sensitivity would be exactly equal to that of the simple descriptive scale.

When assessing the response to analgesics, it is better to use a pain relief score than to measure pain. If pain relief is obtained by subtracting the pain score after treatment from the initial pain score, a relationship will always be found between pain relief and initial pain score [3]. A small imbalance between initial values in two groups of patients may lead to a significant difference in the pain relief obtained. With a pain relief scale, all patients start at the same baseline and all have the same amount of potential response.

When analysing the results of a trial, the distribution of the data should be considered. One of the conditions for the use of the '*t*' test is that the observations are drawn from a normally distributed population. It is safer to treat data from visual analogue and graphic rating scales as ordinal and use non-parametric methods of analysis.

When preparing a scale, the following points should be considered, based on Freyd [1] and Huskisson [4]. Define the sensation or response to be observed. Decide on the extremes of the sensation or response; the end phrases should not be so extremely worded as never to be employed. It is not necessary to alternate the favourable extremes of the line. The descriptive terms should be short, readily understood expressions in common use and to the point. The median of the sensation or response should be in the centre of the line. Numbers should not be superimposed on a visual analogue scale because certain numbers are preferred and interfere with the distribution of results. The line should be of a length that may be grasped as a unit (10 cm is convenient) and should have definite 'cut off' points. The scale should be introduced with an appropriate question, standardised before the experiment.

Experience suggests that prior to a patient using a visual analogue scale at home, it is essential he fills one in under supervision at the clinic. If this is not done, however clear the instructions, many patients will forget what they were told in the clinic and not understand the concept of the scale. In early experiments designed to show the use of a visual analogue scale for analgesic studies, no particular measures were taken to give the patients experience in the use of the scale and failure rates were high. More recently, patients have used visual analogue scales as part of an assessment at the clinic before being asked to use them at home. Using this regime, failures are very unusual.

Because of their sensitivity, visual analogue and graphic rating scales represent the best method for measuring pain or pain relief. Despite developments in measurement techniques, the documentation of pain relief will continue to be important since this represents the object of most treatment. Measurement of pain must always be subjective since pain is a subjective phenomenon — only the patient can therefore measure its severity.

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